Page 1

[Aron] For four hundred thousand years, the Neanderthals survived ice age after ice age.

But their DNA suggests there were never many of them:

Slide notes: Neanderthals lived in Europe and western Asia from about 400,000 to 40,000 years ago and survived many ice ages. Genetic data suggest their population was always small, perhaps only a few thousand breeding adults. Their extinction probably had several causes (climate, disease, competition with modern humans). We use them as a hook: could being few have been enough on its own?

Page 2

[Aron] perhaps only a few thousand breeding adults, spread across all of Europe.

Around forty thousand years ago, they vanished. Climate, disease and competition from us probably all played a part. But one study by Vaesen et al asked a simpler question: could being that few have been enough on its own?

Slide notes: Each dot is one small band of Neanderthals spread across Europe. The positions are illustrative, not real sites. The point is how few and how scattered they were: neighbouring bands could be far apart, which makes finding mates and sharing help harder.

Page 3

[Aron] To see why it might, start with what you'd expect. When a population shrinks, you'd expect the survivors to do better: less competition means more food and space for each one.

Slide notes: The usual expectation (negative density dependence): when a population shrinks, each survivor has more food and space, so it should do better. Each dot is one individual and the circle around it is its share of resources. With few neighbours, each circle is bigger.

Page 4

[Aron] But neighbours aren't just competitors. They're also partners, lookouts and teammates, and losing too many of them may leave each individual worse off.

Slide notes: Neighbours are not only competitors. They are also mates, lookouts and helpers (lines show partners). With too few neighbours, each individual loses that help. This is the idea behind the Allee effect: at low numbers, the cost of losing partners can outweigh the benefit of less competition.

Page 5

[Aron] And that’s what the simulation found: as groups shrank, the birth rate dropped,

Slide notes: Vaesen et al. built a model of small Neanderthal bands that included inbreeding, Allee effects and chance events. In the simulation, as bands shrank, individuals found mates less often and the birth rate dropped.

Page 6

[Aron] shrinking the population further, all the way to extinction.

Slide notes: The lower birth rate shrank the population further, which lowered the birth rate again: a feedback loop that ran to extinction in the model. This shows that small size alone could have been enough. It is a simulation, so it shows what was possible, not what happened.

Page 7

[Zoe] This concept is called the Allee effect: the smaller the population, the lower each individual’s chance of surviving or reproducing.

Slide notes: Definition: the smaller the population, the lower each individual's chance of surviving or reproducing. In other words, a positive relationship between individual fitness and population size or density at low numbers. Named after W.C. Allee, who described it in the 1930s.

Page 8

[Zoe] So why would being few hurt?

First, finding a mate. Spread thin, individuals struggle to meet: for example sea urchins, who fertilise fewer eggs the further apart they are.

Second, safety. A group has more eyes, and a predator can only eat so many, so each animal’s risk is diluted. Vancouver Island marmots, now down to a few scattered animals, spend ten times longer on lookout than they used to, and feed far less.

Third, teamwork. African wild dogs hunt and raise their pups together, and packs of fewer than four adults are thought to be unable to breed successfully.

Fourth, changing their surroundings: bark beetles can only overcome a tree’s defences if enough of them attack at once.

But here’s the catch. Being few also has an upside: less competition. So a population only shrinks if these costs outweigh that benefit, and in the wild, that balance is very hard to measure.

Slide notes:

Four common reasons why being few hurts:

- Mates: spread-out individuals struggle to meet. Sea urchins release eggs and sperm into the water, so fewer eggs are fertilised when urchins are far apart.

- Safety: groups have more eyes and dilute each animal's predation risk. Vancouver Island marmots, now few and scattered, spend much more time on lookout and much less time feeding.

- Teamwork: African wild dogs hunt and raise pups together. Packs of fewer than about four adults are thought to be unable to breed successfully.

- Changing habitat: bark beetles can only overcome a tree's defences if enough attack at once.

Catch: being few also means less competition, so a population only declines if these costs outweigh that benefit.

Page 9

[Zoe] So our question is: are Allee effects an ecological curiosity, or a fundamental driver of population dynamics?

Slide notes: Are Allee effects an ecological curiosity (seen in a few odd species) or a fundamental driver of population dynamics? To answer it we look at the types of Allee effect, then at how strong the evidence is.

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Slide notes: A component Allee effect is "a positive relationship between a specific component of individual fitness and population density". A fitness component is one measurable part of fitness: survival, reproductive success, mate finding, offspring survival. A component effect does not mean the whole population is declining. It only means one part of fitness gets worse at low numbers, and other parts (such as less competition) may make up for it.

Page 11

Slide notes:

How to read it: both charts have pack size on the horizontal axis. Each dot is one pack in one year. Left: pups produced per dog (per capita productivity). Right: percentage of pups that survived.

What it shows: small packs produced fewer pups per dog and fewer of their pups survived. Both measures rose with pack size and peaked at about 11 (productivity) and 12 (pup survival) members. These are component Allee effects, because each is one part of fitness.

Key caveat: the effect was linked to pack size, not total population size. The study found no detectable demographic Allee effect at the population level.

Page 12

Slide notes:

Left (individual level): from the 1970s to 2002-05, colonies shrank from about 8.3 to 3.6 marmots (-57%, our calculation) and time spent feeding fell by about 86%. Survivors also greeted each other

less, heard fewer alarm calls and spent more time watching for predators. Fewer neighbours is the cause; less feeding and more vigilance are behavioural costs that can lower survival and reproduction. They are the route to a component effect, not fitness measures themselves.

Right (population level): each dot is one year, 1970-2007. Horizontal axis: population size. Vertical axis: growth per marmot. The curve (quadratic fit, $R^2 = 0.55$) shows that smaller populations grew more slowly, and at the smallest sizes growth was negative. That is a demographic Allee effect: the costs add up to the whole population's growth rate.

Caveat: these are field data, not an experiment. Predators also increased, so the authors say only reintroductions can prove the cause.

Page 13

Slide notes:

How to read it: each line is one population's size over time. In a weak Allee effect, small populations grow more slowly than larger ones, but growth per individual stays above zero. There is no threshold, so even a tiny population can still grow.

Flour beetle (Tribolium): in Allee's lab experiments, reproduction per female per day was lowest at very low densities and highest at intermediate densities. This is a component measure (one part of fitness). A cost like this can add up to a weak Allee effect at the population level.

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Slide notes:

Left (weak): small populations grow slowly but still grow. Right (strong): there is a critical size, the Allee threshold. Below it, growth per individual is negative (more die than are born), so the population shrinks on its own and goes extinct unless immigrants or management help. Above it, the population grows.

Key difference: weak = lower but positive growth at low numbers; strong = negative growth below a threshold. Only a strong Allee effect creates a point of no return.

Page 15

Slide notes:

The gypsy moth is invasive in North America and spreads by founding small, isolated colonies, where males struggle to find females.

How to read it: horizontal axis = males caught in pheromone traps per colony (a proxy for colony size). Vertical axis = growth from one year to the next, on a log scale: zero means no change, below zero means the colony shrank. The line is the fitted regression from 225 colony-years of trap data; the data points are not shown.

What it shows: the fitted line crosses zero at about 107 males. Below that, colonies are expected to shrink; above it, to grow. This is evidence of a strong Allee effect.

Caveats: about 107 is an estimate from the fitted line, not a sharp cliff. Most data points were at low counts, real colonies scattered widely around the line, chance played a big role, and some colonies below the threshold survived.

Why it matters: to eradicate an invader you do not have to kill every individual, only push colonies below the threshold.

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Slide notes:

Whether Allee effects act as a fundamental driver depends partly on connectivity. In a metapopulation (a network of subpopulations linked by dispersal), a small subpopulation near its Allee threshold can be topped up by immigrants from a large source population. This is the rescue effect, and it can stop a subpopulation from going locally extinct. An isolated subpopulation gets no rescue.

Page 17

Slide ntoes:

A two-patch model with a small patch A and a large patch B. Allee strength (θ) is how strong the Allee effect is; dispersal rate (δ) is how connected the patches are.

How to read it: each panel shows the equilibrium size of each patch (N*) as Allee strength θ increases, at a different dispersal rate (δ = 0, 0.015, 0.075, 0.17). θc is the critical Allee strength: above it, an isolated patch dies out.

What it shows: just above the small patch's critical strength, more connectivity helps a lot, because the big patch rescues the small one. But dispersal also drains individuals from patches, so stronger rescue can lower the Allee strength at which the whole metapopulation goes extinct. Connectivity can help or hurt.

Page 18

Slide notes:

How to read it: each panel shows total population size (vertical) against dispersal rate δ (horizontal) for one fixed Allee strength θ, from weak (E, θ = 0.05) to strong (K, θ = 0.7). Dashed line: no dispersal.

What it shows: at weak Allee strength, more dispersal always helps (MB, monotonically beneficial). As θ rises, an "Allee pit" appears first: at low-to-moderate dispersal, the small patch loses too many individuals and falls below its Allee threshold without enough rescue, so the total population drops below the no-dispersal level. At higher θ, dispersal turns from beneficial to harmful (BTD, MD), and at the strongest θ the whole population goes extinct (K).

Key point: immigration does not always rescue. Whether connectivity helps depends on how strong the Allee effect is and how fast individuals move.

Page 19

Slide notes:

Field data from the Aland Islands metapopulation in Finland. Population size is measured in larval groups (each typically 50-100 larvae before overwintering diapause).

How to read it: left = fraction of populations that increased; right = new larval groups produced per group. Bars compare isolated and connected populations; error bars are mean ± 1 SE, pooled over three annual transitions (1994-96).

What it shows: the smallest populations (one larval group) did worse on both measures when isolated. In larger populations, isolated and connected did not differ significantly. So the Allee effect showed up only in small, isolated populations, and connectivity (immigration) hid it. Without looking at connectivity, the authors would not have detected the Allee effect.

Page 20

Slide notes:

Setup: tanks of water fleas (Daphnia) at different starting sizes, plus a predator (a midge larva) that can only eat so fast. Because the predator's intake is capped, each flea in a small population is far more likely to be eaten.

Result: small populations died out faster, a predator-driven Allee effect causing extinction.

Strength: shows the mechanism clearly and controls density. Blind spot: only in a tank, so it may not reflect wild populations.

Page 21

Slide notes:

Setup: island fox populations on separate Channel Islands, California, used as natural replicates. Numbers had crashed, largely because of golden eagles.

Result: in small populations two costs acted at once (a "double Allee effect"): higher per-capita eagle predation and lower mate finding. But there was also a benefit: less competition meant bigger litters and better pup survival. Overall the costs won, so small populations grew more slowly: a weak Allee effect.

Strength: real populations. Blind spot: the net effect depends on how the separate components are weighed, which is hard to untangle.

Page 22

Slide notes:

Setup: Atlantic cod on St Pierre Bank, northwest Atlantic. Fifty years of stock and recruitment records (1959-2008). Stocks collapsed under fishing and have not recovered after 30 years of reduced fishing.

How to read it: each dot is one year. Horizontal axis = spawning stock biomass (tonnes of adults). Vertical axis = recruits per tonne of spawning stock, i.e. young produced per unit of adults.

Result: below roughly 90,000 t of spawning stock, recruits per tonne fell: adults produced fewer young per unit of stock. This is evidence of an Allee effect.

Blind spot: the evidence only became strong after the crash, because you cannot measure low-density dynamics until a population gets there. You often find the threshold after crossing it.

Page 23

[Zoe] Our assigned classic and modern papers are 25 years apart.

Allee spotted the effect in 1931, but Stephens and colleagues, in 1999, pinned down what actually counts: the split between component and demographic effects we saw earlier.

In 2024, Muir and colleagues combined 62 experiments and found the effect is real. But both papers describe the same gap. Theory is about whole populations, and the experiments mostly measure costs to individuals.

Slide notes:

Stephens et al. 1999 (classic) defined the terms, with no new data. Their key distinction is component (one fitness component) vs demographic (whole-population growth) Allee effects.

Muir et al. 2024 (modern) pooled 62 experimental studies in a meta-analysis. They found an overall positive Allee effect whose strength varied by mechanism (such as mate finding), not by taxonomic group. Some mechanisms, such as group living and predation, were not detectable on their own.

Both papers describe the same gap: theory is mostly about demographic effects, but experiments mostly measure component effects, i.e. costs to individuals rather than whole populations.

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[Zoe] So how strong is the evidence for whole populations? Those 62 experiments are the most controlled of the tests.

In the field, 63 of 91 studies found some sign of an Allee effect, but only 7 found a point of no return.

And across almost twelve hundred long-term records of wild populations, an Allee model fit best in just 13.

In a lab, you choose the density and measure one thing, like fertilised eggs. In the wild, you have to show the whole population grows slower when small, despite weather, predators and chance. And proving a point of no return is hardest of all because you'd have to catch a population falling past it.

So the stricter the test, the rarer the evidence.

Slide notes:

How to read it: moving from left to right, each test is stricter (harder to pass).

- Experiments: 62 studies pooled (Muir et al. 2024). Clear effects, but mostly costs to individuals.
- Field studies, any clear evidence: 63 of 91 (69%) (Kramer et al. 2009).
- Field studies showing a point of no return (strong Allee effect): 7 of 91 (8%) (Kramer et al. 2009).
- Wild population records where an Allee model fit best: 13 of 1,198 (1%) (Gregory et al. 2010).

Takeaway: the closer a test gets to whole wild populations and to a threshold, the rarer the evidence. That does not mean Allee effects are rare: weather, predators and chance make them hard to detect, and a threshold can only be proven once a population falls past it. These are separate datasets, so the numbers are not directly comparable.

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So Allee effects are hard to see: costs to individuals don’t always reach the population, weak effects leave no tipping point, and newcomers can hide them. But where they act, they can decide survival, in small, declining, harvested populations like cod, and new arrivals like the gypsy moth.

Slide notes:

Allee effects are hard to see: component costs do not always reach the population, weak effects have no tipping point, and immigrants can hide them. But where they act, they can decide survival. The dashed line is the point of no return of a strong Allee effect.

- Very small populations: chance alone can tip them below the threshold.

- Declining populations: they can slide past it.

- Harvested populations: may stay low after harvesting stops (as in cod).

- Newly arrived populations: may die out before establishing, which is good news for controlling invasive pests (as in the gypsy moth).

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[Aron] Back to the Neanderthals: a small population, scattered across Europe, and possibly cut off from each other as modern humans spread between their groups, so no one could come to the rescue. Those are exactly the conditions where Allee effects decide the outcome.

Slide notes: Neanderthals had all the risk conditions: a small population, scattered across Europe, and possibly cut off from each other as modern humans spread between their groups, so no immigrants could rescue them. These are the conditions where Allee effects are most likely to decide the outcome.

Page 27

[Aron] But we can't prove that's what happened to them, and that's the point.

Slide notes: We cannot prove an Allee effect caused Neanderthal extinction. The model shows it was possible, not that it happened. This mirrors the evidence problem: Allee effects are hardest to demonstrate exactly when they matter most.

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[Aron] So the answer to our question: Allee effects are not a curiosity, but a fundamental driver. They decide the fate of small, declining and harvested populations, often before we can even detect them.

(pause)

Thank you. We’re happy to take questions.

Slide notes: Our answer: not a curiosity, but a fundamental driver. Allee effects are found across many taxa (Muir et al. 2024) and decide the fate of small, declining, harvested and newly arrived populations, often before we can detect them. The evidence is strongest for individual costs and weakest for population thresholds, which reflects how hard they are to measure, not that they are rare.

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